Solve Math Problems... with Coloring!

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Topics I Learned in DRP

- Pigeonhole Principle
- Generating Functions
- Double Counting
- Recurrence Methods
- Coloring Methods
- Combinatorial Counting
- Existence and Inequalities

Overview

Topics we will cover in this presentation:

- Pigeonhole Principle
- Pigeonhole Principle with Graph Coloring
- Coloring Methods for Harder Problems

What is Pigeonhole?

Before we apply the Pigeonhole Principle to anything, we shall first state it.

Statement

If m objects are put into n boxes, then at least one box contains $\left\lfloor \frac{m-1}{n} \right\rfloor + 1$ or more objects, where $\lfloor x \rfloor$ denotes the greatest integer less than or equal to a real number x.

Intuition: If we had some number of boxes, in order to minimize the number of objects in the box with the most objects, we want to make each box have as close to the same number of objects as possible. We use $m - 1$ instead of m because if m were divisible by n, then each box could have the same number of objects, but adding 1 to m would require another object to be in one of the boxes.

Pigeonhole Principle with Graph Coloring

To illustrate the concept of using the pigeonhole principle with graph coloring, we introduce and solve the following problem.

Problem

There are six points in a 3-dimensional space, no four points of which are coplanar. Two distinct points determine a line segment, and let each line segment be colored red or blue. Prove that there exists a triangle such that the 3 sides of this triangle are all colored red or blue.

Pigeonhole Principle with Graph Coloring

Problem

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Brainstorming: *We are faced with a problem that involves a lot of vertices, but only two colors. We just covered the Pigeonhole Principle, so what if we tried to use it here? Instead of thinking about all six points at the same time, we could try to fix one point, A. Then, there would be five points left, each connected to A by a line segment. Five line segments... two colors...*

Solution

Fix one of the points and label it A. Then, there are five points left, each connected to A by a line segment. With five line segments and two colors, we must have that at least $\lfloor \frac{5-1}{2} \rfloor + 1 = 3$ line segments have the same color. (Think of the line segments as objects and the colors as the boxes!) Without loss of generality, suppose this common color were red. We shall refer to all points connected to A with a red line segment as *special*. Since we want to avoid forming a triangle whose sides are all the same color, we decide to color all line segments connecting any two special points blue instead of red. However, these line segments will themselves form a triangle all of whose sides are colored blue. ■

Here is a different type of coloring problem.

Problem

When two adjacent unit squares are removed from a row of a 2×3 rectangle, the resulting shape is called an "L-shape". Given an 8×8 chessboard, is it possible to obtain a *perfect cover* of this chessboard using seven 2×2 squares and nine "L-shapes"?
Note: A *perfect cover* of the chessboard with shapes is one whose shapes do not overlap and whose union (total area) exactly spans the area of the chessboard.

Problem

Given an 8×8 chessboard, is it possible to obtain a *perfect cover* of this chessboard using seven 2×2 squares and nine "L-shapes"?

Brainstorming: *Let's try coloring each square of the chessboard with one of two colors: black and white. But how? Maybe we could color in such a way that the two types of shapes are colored differently. Maybe we should consider the number of black and white squares covered by each shape... but we need a coloring that can actually be useful.*

Solution

Color every row the same color, alternating between black and white. There will then be 32 black and 32 white squares in the chessboard. Observe that any 2×2 square will cover two black and two white squares, and any "L-shape" will cover either one black and three white squares, which we will call *W-shapes*, or one white and three black squares, which we will call *B-shapes*. Let x denote the number of B-shapes; there will then be $9 - x$ W-shapes. In total, we have

$$(2 \cdot 7) + (3 \cdot x) + (1 \cdot (9 - x))$$

black squares covered by the seven 2×2 squares and the nine "L-shapes".

(Continued on the next slide.)

Solution

However, setting this expression equal to 32, the total number of black squares in the chessboard, we get

$$(2 \cdot 7) + (3 \cdot x) + (1 \cdot (9 - x)) = 32,$$

which yields $x = \frac{9}{2}$. As x must be an integer, this shows that a perfect cover with the given shapes is not attainable. ■

Conclusion

tl;dr

Coloring is hard!

When dealing with problems that involve graphs or boards, namely connections and coverings, coloring will likely come in use. However, the challenge arises when we have to think of how to color such that we can form clear contradictions to solve problems.

References

Yao Zhang.
Combinatorial Problems in Mathematical Competitions.
East China Normal University Press, 2011.